Sequence 8

ALGEBRA, EQUATIONS AND INEQUALITIES

4–20 January 2027

I Brackets and distributivity

Removing brackets

After a + sign, removing the brackets keeps the signs of the terms. After a − sign, every term inside changes sign. This rule applies to a sum or a difference of terms.

Example 8.1

Remove the brackets and simplify: $A = x + (2 + 3x)$; $B = x - (2 + 3x)$; $C = -2x + (-4 + 5x)$; $D = -2x - (-4 + 5x)$.

Solution:

$A = 4x + 2$; $B = -2x - 2$; $C = 3x - 4$; $D = -7x + 4$.

Distributivity

For all real numbers a, b, c, d:

$$b(c + d) = bc + bd, \qquad (a + b)(c + d) = ac + ad + bc + bd.$$

Example 8.2

Expand and simplify:

$$A = -4(8x + 7) + (-8x + 5)(5x + 9),$$
$$B = (6x + 9)(-x + 7) + (3x + 5)(-9x + 6),$$
$$C = -9(-x - 7) - (-x + 3)(-8x - 8),$$
$$D = -5(-6x - 7) + (x - 7)(5x + 8).$$

Solution:

Expand each product, then collect terms of the same degree:

$$A = -32x - 28 - 40x^2 - 72x + 25x + 45 = -40x^2 - 79x + 17,$$
$$B = -6x^2 + 33x + 63 - 27x^2 - 27x + 30 = -33x^2 + 6x + 93,$$
$$C = 9x + 63 - (8x^2 - 16x - 24) = -8x^2 + 25x + 87,$$
$$D = 30x + 35 + 5x^2 - 27x - 56 = 5x^2 + 3x - 21.$$

II Factorising with a common factor

Property

$ka + kb = k(a + b)$ and $ka - kb = k(a - b)$. Factorising turns a sum into a product; expanding does the reverse.

Example 8.3

Factorise:

$$A = 15x^2 - 21x,$$
$$B = 3(2 + 3x) - (5 + 2x)(2 + 3x),$$
$$C = (4x - 1)(x + 6) + (4x - 1),$$
$$D = (1 - 6x)^2 - (1 - 6x)(2 + 5x).$$

Solution:

$A = 3x(5x - 7)$.
$B = (2 + 3x)[3 - (5 + 2x)] = (2 + 3x)(-2 - 2x) = -2(2 + 3x)(1 + x)$.
$C = (4x - 1)[(x + 6) + 1] = (4x - 1)(x + 7)$.
$D = (1 - 6x)[(1 - 6x) - (2 + 5x)] = (1 - 6x)(-1 - 11x) = (6x - 1)(11x + 1)$. Check the factors by expanding again.

Example 8.4

Choose a useful form of $f(x) = (x - 2)(x + 3)$ to calculate $f(2)$, $f(0)$ and find its zeros.

Solution:

The factorised form immediately gives $f(2) = 0$ and the zeros $2, -3$. At 0, $f(0) = (-2) \times 3 = -6$; the expanded form $x^2 + x - 6$ also gives this result. The choice depends on the question.

III Special identities: expanding

Identities

$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a-b)^2 = a^2 - 2ab + b^2,$$
$$(a+b)(a-b) = a^2 - b^2.$$

Proof to write.

Solution:

Expand $(a+b)(a+b)$: $a^2+ab+ba+b^2 = a^2+2ab+b^2$. Replacing b by $-b$ gives the second identity. Finally $(a+b)(a-b) = a^2 - ab + ab - b^2 = a^2 - b^2$.

Example 8.5

Expand $(x+3)^2$, $(3x-4)^2$, $(x-3)(x+3)$, $(2x+3)(2x-3)$, $(4-3x)(3x+4)$.

Solution:

x^2+6x+9; $9x^2-24x+16$; x^2-9; $4x^2-9$; $16-9x^2$. In the last product, reorder $3x+4$ as $4+3x$.

Example 8.6

Expand and simplify $A = (2x-3)^2 + (x+5)(3-x)$ and $B = 2(x+3) + (2x+3)(2x-3)$.

Solution:

$A = 4x^2 - 12x + 9 - x^2 - 2x + 15 = 3x^2 - 14x + 24$. $B = 2x + 6 + 4x^2 - 9 = 4x^2 + 2x - 3$.

IV Special identities: factorising

Method

Spot the two squares; for three-term forms, check the double product $2ab$. For a difference of two squares, use $a^2 - b^2 = (a - b)(a + b)$.

Example 8.7

Factorise:

$$A = x^2 - 2x + 1, \quad B = 25 + 16x^2 - 40x, \quad C = 4x^2 + 12x + 9,$$
$$D = 1 - 49x^2, \quad E = 9x^2 - 4, \quad F = (2x + 3)^2 - 64.$$

Solution:

$A = (x - 1)^2$. $B = (4x - 5)^2$. $C = (2x + 3)^2$.
$D = (1 - 7x)(1 + 7x)$. $E = (3x - 2)(3x + 2)$.
$F = [(2x + 3) - 8][(2x + 3) + 8] = (2x - 5)(2x + 11)$. For the first three, check the middle term; for F, treat $2x + 3$ as a single block.

Example 8.8: Checking an identity

Are the equalities $(a + b)^2 = a^2 + b^2$ and $a^2 + b^2 = (a - b)(a + b)$ true for all real numbers?

Solution:

No: for $a = b = 1$, the first would give $4 = 2$ and the second $2 = 0$. The missing term in the first is $2ab$; the product in the second equals $a^2 - b^2$.

Fluency – 5 to 10 minutes

Calculate 101^2 mentally; give a direction vector of $2x + 3y = 1$; determine $[-1; 3] \cap [2; 5]$.

Answers: 10201; $(-3; 2)$; $[2; 3]$.

V Integer powers

Definition and rules

For $n \geq 1$, a^n is the product of n factors equal to a. For $a \neq 0$, $a^0 = 1$ and $a^{-n} = 1/a^n$. In this course, 0^0 is not defined. For $a, b \neq 0$ and integers m, n:

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

For strictly positive exponents, some bases may be zero if the expressions remain defined.

Example 8.9

Simplify $A = 2^4 \times 2^3$, $B = (-5)^{12}/(-5)^3$, $C = (7^3)^6 \times 7^{-5}$, $D = 7^{24} \times 3^2 \times 2^3$. Also calculate $(-3)^5$, 5^1 and 99999^0.

Solution:

$A = 2^7 = 128$. $B = (-5)^9 = -1953125$. $C = 7^{18-5} = 7^{13}$. $D = 72 \times 7^{24}$: exponents of different bases cannot be added. The last three values are -243, 5, 1.

Watch out. The definition as a repeated product does not hold for a negative exponent. $-3^2 = -9$, whereas $(-3)^2 = 9$.

Example 8.10

Solve $2^n > 100$ mentally for the smallest natural number n. Write a program that finds it.

Solution:

$2^6 = 64 \leq 100 < 128 = 2^7$, so $n = 7$. Initialise $n = 0$, $p = 1$; while $p \leq 100$, replace p by $2p$ and n by $n + 1$. The program is given at the end of the sequence.

VI Fractions and fractional expressions

Rules and conditions

For nonzero denominators:

$$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}, \quad \frac{a}{c} \pm \frac{b}{d} = \frac{ad \pm bc}{cd}, \quad \frac{a}{c} \times \frac{b}{d} = \frac{ab}{cd}.$$

A fraction c/d has an inverse only if $c \neq 0$ and $d \neq 0$. Then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc} \quad (b, c, d \neq 0).$$

Example 8.11

Calculate and reduce to lowest terms:

$$A = \left(\frac{2}{3} + \frac{1}{6}\right) \times \frac{4}{5}, \quad B = \frac{1}{2} + \frac{3}{4} \times \frac{5}{6}, \quad C = \frac{4}{5} \div \frac{2}{3} \times \frac{7}{10},$$

$$D = \frac{\left(\frac{2}{5} + \frac{1}{3}\right) \times \frac{3}{4}}{\frac{1}{2} - \frac{1}{6}}.$$

Solution:

$A = \frac{5}{6} \times \frac{4}{5} = \frac{2}{3}$. $B = \frac{1}{2} + \frac{5}{8} = \frac{9}{8}$. $C = \frac{4}{5} \times \frac{3}{2} \times \frac{7}{10} = \frac{21}{25}$. $D = \frac{11}{15} \times \frac{3}{4} \div \frac{1}{3} = \frac{11}{20} \times 3 = \frac{33}{20}$. In C, multiplication and division are carried out from left to right.

Example 8.12

Simplify $\dfrac{x^2 - 1}{x - 1}$, stating the domain. Can we then take $x = 1$?

Solution:

For $x \neq 1$, $\frac{(x-1)(x+1)}{x-1} = x + 1$. The original expression remains undefined at 1; the simplification does not lift the restriction.

VII Square roots and validity conditions

Definition and properties

For $a \geq 0$, $\sqrt{a}$ is the unique positive or zero number whose square is a. Thus $(\sqrt{a})^2 = a$. For every $x \in \mathbb{R}$, $\sqrt{x^2} = |x|$. For $a, b \geq 0$, $\sqrt{ab} = \sqrt{a}\sqrt{b}$. If $b > 0$, $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$.

Proof to write.

Solution:

The number $\sqrt{a}\sqrt{b}$ is positive or zero and its square is ab; by uniqueness of the positive square root, it equals $\sqrt{ab}$. The same argument applied to the quotient establishes the second formula, under $b > 0$. Similarly $|x| \geq 0$ and $|x|^2 = x^2$, so $\sqrt{x^2} = |x|$.

Example 8.13

Simplify $\sqrt{18}$; $\sqrt{2}\sqrt{8}$; $\sqrt{7/25}$; $\sqrt{27/75}$; $\sqrt{(-5)^2}$.

Solution:

$3\sqrt{2}$; 4; $\sqrt{7}/5$; $\sqrt{9/25} = 3/5$; 5.

Watch out. There is no general rule $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$. With $a = 4, b = 9$, we would wrongly have $\sqrt{13} = 5$. Saying these expressions are always different would also be wrong, for example if $a = 0$.

Example 8.14

Simplify $2\sqrt{18} - \sqrt{50} + \sqrt{8}$. Bound $\sqrt{20}$ between two consecutive integers.

Solution:

$6\sqrt{2} - 5\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$. Since $16 < 20 < 25$, $4 < \sqrt{20} < 5$.

VIII First-degree equations

Definition

To solve an equation is to find all the values of the unknown that satisfy the equality. Two equations are equivalent if they have the same solutions. Adding the same expression or multiplying by the same nonzero number preserves equivalence.

Method

Expand if necessary, collect the terms in x, then isolate x. If its coefficient is zero, examine the remaining equality: it may be always true or impossible. Check in the original equation.

Example 8.15

Solve:

$$a)\ 5x-3=2x+9, \qquad b)\ 3(2x-1)=2(x+4)+7,$$
$$c)\ -2(x+3)=4x-(x-1), \qquad d)\ \frac{x+2}{3}=x-1,$$
$$e)\ \frac{-5x+5}{8}=-\frac{3}{4}, \qquad f)\ \frac{-3x-1}{9}=-\frac{7}{4}.$$

Solution:

a) $3x=12$, so $x=4$. b) $6x-3=2x+15$, so $x=9/2$. c) $-2x-6=3x+1$, so $x=-7/5$. d) $x+2=3x-3$, so $x=5/2$. e) $-5x+5=-6$, so $x=11/5$. f) $4(-3x-1)=-63$, so $-12x=-59$ and $x=59/12$.

Example 8.16

Solve $2(x+1)=2x+2$, then $2(x+1)=2x+3$.

Solution:

The first reduces to $2=2$: $S=\mathbb{R}$. The second gives $2=3$: $S=\varnothing$. We do not divide by a zero coefficient.

IX Zero-product property and equations with a quotient

Properties

$AB = 0 \iff A = 0$ or $B = 0$. For $B \neq 0$, $A/B = 0 \iff A = 0$. To solve $A(x)/B(x) = k$, first require $B(x) \neq 0$, then solve $A(x) = kB(x)$ and discard the forbidden values.

Example 8.17

Solve $(4x + 6)(3 - 7x) = 0$; $3x(-7x + 5) = 0$; $\dfrac{2x + 1}{3x - 5} = 0$.

Solution:

Solutions: $\{-3/2; 3/7\}$; $\{0; 5/7\}$; $\{-1/2\}$, with forbidden value $5/3$ in the last case.

Example 8.18

Factorise then solve $(1 - 6x)^2 - (1 - 6x)(2 + 5x) = 0$ and $(4x - 1)(x + 6) + (4x - 1) = 0$.

Solution:

First: $(6x - 1)(11x + 1) = 0$, so $S = \{1/6; -1/11\}$. Second: $(4x - 1)(x + 7) = 0$, so $S = \{1/4; -7\}$.

Example 8.19

Solve $\dfrac{x + 1}{x - 2} = 3$, then $\dfrac{x^2 - 4}{x - 2} = 4$.

Solution:

In both cases, $x \neq 2$. First: $x + 1 = 3x - 6$, so $x = 7/2$, allowed. Second: for $x \neq 2$, the quotient equals $x + 2$; the equation requires $x = 2$, forbidden. So no solution.

X Equations with a square; isolating a variable

Reminder

In $\mathbb{R}$, $x^2 = a$ has two solutions $\pm\sqrt{a}$ if $a > 0$, one solution 0 if $a = 0$, and none if $a < 0$.

Example 8.20

Solve $x^2 = 0$; $x^2 = -8$; $x^2 = 7$; $25x^2 = 81$; $(x+2)^2 = 9$.

Solution:

In order: $\{0\}$; $\varnothing$; $\{-\sqrt{7}; \sqrt{7}\}$; $\{-9/5; 9/5\}$; $\{-5; 1\}$. For the last one, $x + 2 = -3$ or $x + 2 = 3$.

Isolating a variable in a formula

Describe the operations performed on both sides; state the conditions that allow dividing or taking a square root.

Example 8.21

Isolate h in $V = \pi r^2 h$ with $r > 0$; isolate v in $E = \frac{1}{2}mv^2$ with $m > 0$, $E \geq 0$ and $v \geq 0$; isolate x in $y = ax + b$ with $a \neq 0$.

Solution:

$h = V/(\pi r^2)$. Then $v^2 = 2E/m$; since $v \geq 0$, $v = \sqrt{2E/m}$. Finally $x = (y - b)/a$. Without the condition $v \geq 0$, two signs would be possible for $E > 0$.

Fluency – 5 to 10 minutes

Give the domain of $1/(x+3)$; calculate $\sqrt{49}$; the slope between $(0; 2)$ and $(3; 8)$.

Answers: $\mathbb{R} \setminus \{-3\}$; 7; 2.

XI Inequalities and comparison

Rules

Inequalities of the same direction can be added: $a \leq b$, $c \leq d \Rightarrow a + c \leq b + d$. Adding the same real number preserves the direction. Multiplying or dividing by a strictly positive real number preserves the direction; by a strictly negative one, it reverses it. Comparing A and B amounts to studying the sign of $A - B$. If $A, B > 0$, one can also compare A/B to 1.

Example 8.22

1. If $x \leq -4$, what can be said about $3x - 4$? 2. If $x > -6$, what can be said about $2x - 3$? 3. If $7 \leq x \leq 16$ and $3 \leq y \leq 12$, bound $8x + 7y$. 4. If $-10 \leq x \leq -6$, bound $2x - 6$.

Solution:

1. $3x - 4 \leq -16$. 2. $2x - 3 > -15$. 3. $56 \leq 8x \leq 128$ and $21 \leq 7y \leq 84$, so $77 \leq 8x + 7y \leq 212$. The bounds are attainable. 4. $-26 \leq 2x - 6 \leq -18$.

Example 8.23

Solve $3x + 1 \geq x - 7$; $2x - 11 \leq 4x - 5$; $6x + 19 \leq 2x - 3$.

Solution:

1. $2x \geq -8$, that is $[-4; +\infty[$. 2. $-6 \leq 2x$, that is $[-3; +\infty[$. 3. $4x \leq -22$, that is $]-\infty; -11/2]$.

XII Sign of an affine expression and of a product

Property

For $a \neq 0$, $ax + b$ vanishes at $x_0 = -b/a$; its sign to the right is that of a, to the left the opposite. If $a = 0$, the sign is that of the constant b. For a product, multiply the signs of the factors.

Example 8.24

Draw the sign tables of $3 - x$, $2x - 6$, then $P(x) = (2x - 6)(x + 1)$.

Solution:

Both affine expressions vanish at 3: $3 - x$ is positive on the left, negative on the right; $2x - 6$ has the opposite signs.

x	$-\infty$		-1		3		$+\infty$
$2x - 6$		$-$	$-$	$-$	0	$+$	
$x + 1$		$-$	0	$+$	$+$	$+$	
$P(x)$		$+$	0	$-$	0	$+$	

Example 8.25

Solve $(7 + 3x)(2x + 6) > 0$.

Solution:

The zeros are $-7/3$ and -3, to be ordered $-3 < -7/3$. The product is positive outside these roots and negative between them. The inequality is strict: $S =]-\infty; -3[\cup]-7/3; +\infty[$.

XIII Sign of a quotient and algorithmic synthesis

Method

Find the forbidden values, then the zeros of the numerator. Order all the critical values and apply the sign rule. On the quotient row, a double bar indicates a forbidden value.

Example 8.26

Draw the sign table of $Q(x) = \dfrac{-4x+6}{x-3}$, then solve $\dfrac{2x-1}{-4x+3} \leq 0$.

Solution:

For Q, zero $3/2$, forbidden 3. Signs: negative on $]-\infty;3/2[$, zero at $3/2$, positive on $]3/2;3[$, undefined at 3, negative on $]3;+\infty[$. For the second expression, zero $1/2$, forbidden $3/4$. The signs are $-,+,-$ on the three intervals. Thus $S =]-\infty;1/2] \cup]3/4;+\infty[$.

```
def premiere_puissance(seuil):
    # seuil reel >= 1
    n, p = 0, 1
    while p <= seuil:
        n = n+1
        p = 2*p
    return n, p
```

Example 8.27

What does `premiere_puissance(100)` return? What relation between `p` and `n` stays true at each step? How can the sign of a number then be tested in Python?

Solution:

The result is $(7, 128)$. The invariant is $p = 2^n$, true at initialisation and then preserved. For the sign, first test `x > 0`, then `x < 0`, otherwise $x = 0$. Do not replace the loop's non-strict inequality by a strict one without changing the problem posed.